MILITARY SPECIFICATION

CYLINDER, NITROGEN, FOR RESCUE HAND TOOL

This specification has been approved by the
Naval Air Systems Command, Department
of the Navy

1 SCOPE

1.1 Scope - This specification covers a puncturable seal type nitrogen gas filled cylinder.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-6998	Dichloromethane, Technical

STANDARDS

Military

MIL STD-105	Sampling Procedures and Tables for Inspection by Attributes

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STANDARDS

Military (Cont'd)

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality assurance representative. Copies of this specification and other unclassified specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

3.1 First article - This specification makes provisions for first article inspection.

3 2 Materials - Materials used in the construction of the cylinder shall be in accordance with the applicable drawings and this specification

3 2 1 Nitrogen - The nitrogen used in the cylinders shall conform to BB-N-411, Type I, Grade B, Class 1

3 3 Construction - The cylinders shall be constructed in accordance with Figure 1

3 4 Performance -

3 4 1 Cleanliness - The cylinder, when tested as specified in 4 6 1.1, shall pass the requirements specified in Table II

3.4.2 Gas charge - The cylinder, when tested as specified in 4.6.2, shall:

(a) not show any evidence of deformation.

(b) weigh not less than 9 grams nor more than 10 grams at 3000 psig.

3.4.3 Leakage - The cylinder, when tested as specified in 4.6.3, shall not show any evidence of gas bubbles after 1 minute of submergence.

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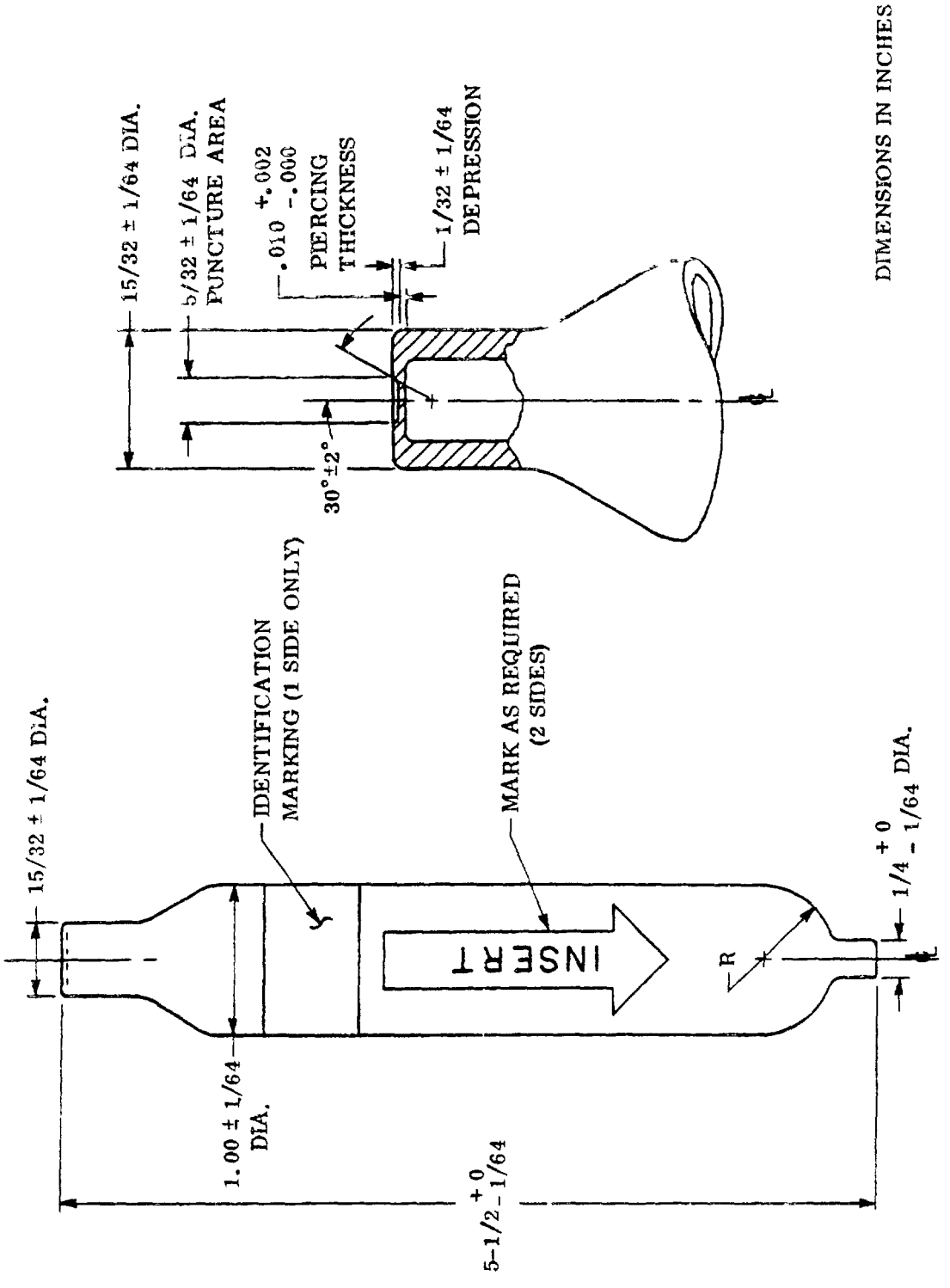


FIGURE 1. NITROGEN GAS CYLINDER

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3.4.4 Cyclic leakage - The cylinder, when tested as specified in 4.6.4, shall not show any evidence of leakage

3.4.5 Hydrostatic pressure - The cylinder, when tested as specified in 4.6.5, shall not show any evidence of bursting

3.4.6 Rupture - The cylinder, when tested as specified in 4.6.6, shall not show any evidence of splintering or fragmentation Complete separation of the puncturable area or any portion of the cylinder shall be considered as fragmentation

3.4.7 Maximum filling density - The cylinder, when tested as specified in 4.6.7, shall have a maximum filling density of 24 percent The filling density is the percent ratio of the weight of nitrogen in a container to the weight of water that container will hold The maximum filling density shall be determined as follows.

$$\text{Filling density} = \frac{\text{Weight of } N_2}{\text{Weight of } H_2O} \times 100$$

3 5 Finish - The surface of the cylinder shall be cleaned in accordance with Method 1 of TT-C-490 and cadmium plated in accordance with the requirements for Type I, Class C of QQ-P-416

3 6 Identification of product - The cylinder shall be marked for identification in accordance with MIL-STD-130 The cylinder shall be marked by ink stamping with letters not less than 1/4 inch high to indicate the following information

Cylinder, Nitrogen, 3000 PSI, for Hand Power Tool
MIL-C-81575
Manufacturer Identification
Date of Manufacture
Gross Weight Grams

A directional arrow, as shown in Figure 1, with the word "INSERT" as a guide, shall be ink stamped, with letters not less than 1/8 inch high, along the body of the cylinder and to the left of the information stated above

3 6 1 Indicated gross weight shall be the empty weight of the cylinder (rounded off to the next higher 1/2 gram) plus the weight of the minimum gas charge

3 7 Workmanship - The cylinders shall be uniform in quality and shall be free from irregularities or defects which could adversely affect safety, performance, reliability, or durability The cylinders shall be free of oil, grease, fuel, water, dust, dirt, or any other foreign matter

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the cylinders shall be classified as follows:

(a) First article inspection - First article inspection shall consist of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.3).

(b) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4).

4.3 First article inspection - The first article inspection of the cylinders shall consist of examinations and tests for all the requirements of this specification.

4.3.1 First article inspection samples - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit 4 cylinders. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these cylinders from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiving of the preproduction inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article samples shall be furnished Government as directed by the contracting officer (see 6.2).

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4.3.2 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. Two approved cylinders will be returned to the manufacturer for use in monitoring production. The other cylinders will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.4 Quality conformance inspection - The quality conformance inspection shall consist of the following tests and examinations:

Visual examination of the filled cylinders
Dimensional examination of the filled cylinders
Gas Charge
Leakage
Deformation
Cyclic leakage
Hydrostatic pressure
Rupture
Maximum filling density
Preparation for delivery

4.4.1 Sampling

4 4 1 1 Inspection lot -

4 4.1 1.1 Cylinder - An inspection lot size shall be expressed in units of one cylinder made under essentially the same conditions and from the same materials The sample unit shall be one cylinder

4.4 1 1 2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing cylinders fully prepared for delivery, made from essentially the same materials The sample unit shall be one shipping container, containing cylinders fully prepared for delivery with the exception that it need not be sealed

4 4 1 2 Sampling for tests and examinations of cylinders - The sample size, acceptance criteria, tests, and examinations required for the cylinders shall be as specified in Table I

4 5 Test conditions -

4 5.1 Gas - Unless otherwise specified, the gas used in testing shall be nitrogen conforming to BB-N-411, Type 1, Class 1, Grade B

4.5.2 Cleanliness solvent - The solvent used in the cleanliness test shall be dichloromethane conforming to MIL-D-6998

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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATION OF THE CYLINDERS

INSPECTION	METHOD PARAGRAPH	SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
Visual examination (see Classification of Defects)	4.6.1.1	Every cylinder for critical defects. Inspection Level II 1/ for minor defects.	Reject all units with any critical defects. An Acceptable Quality Level of 2.5 defects per hundred units for minor defects.
Dimensions	4.6.1.1.1	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Gas charge	4.6.2	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Leakage	4.6.3	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Cyclic leakage	4.6.4	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Hydrostatic pressure	4.6.5	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Rupture 2/	4.6.6	Inspection Level S-2 2/	An Acceptable Quality Level of 2.5 defects per 100 units
Maximum filling density	4.6.7	Inspection Level S-3	An Acceptable Quality Level of 2.5 defects per 100 units
Examination of preparation for delivery	4.6.1.2	Inspection Level S-2	An Acceptable Quality Level of 4.0 defects per 100 units

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

2/ These are destructive tests (see 6.2).

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4 5 3 Temperature and pressure - Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions NTP conditions are 29 92 inches of mercury and 70° F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4 6 Inspection methods -

4 6 1 Visual examination -

4 6 1 1 Cylinder - Every cylinder shall be examined visually (for critical defects) to determine conformance to this specification. The classification of defects, Table II, shall be used to classify the defects found

4.6.1.1.1 Dimensions - The cylinders shall be checked dimensionally to determine conformance to the dimensions specified in applicable drawing.

4 6 1 2 Preparation for delivery - Each of the fully prepared shipping containers, containing cylinders selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification The classification of defects, Table III, shall be used to classify the defects found

4.6.2 Gas charge - Each cylinder shall be weighed prior to charging with nitrogen. The cylinder shall then be charged with nitrogen, conforming to BB-N-411, Type I, Grade B, Class 1, at 3000 psig and then heated to a temperature of plus 160 ±5° Fahrenheit, and maintained at this temperature for 20 minutes. After the 20 minute period the cylinder shall meet the requirements of 3.4.2 (a). After a period of 24 hours the cylinder shall meet the requirements of 3.4.2 (b).

4.6.3 Leakage - The cylinder shall be charged to 3000 psig. With the pressure maintained the cylinder shall be immersed in a water solution (1 percent aerosol OT clear, 25 percent or equal) to a depth so that the uppermost portion of the cylinder is at least 1 inch below the water level for a period of two minutes. A 20-inch vacuum (Mercury) shall be maintained above the surface of the solution and the cylinder shall pass the requirements specified in 3.4.3.

4.6.4 Cyclic leakage - Filled cylinders (3000 psig) shall be subjected to two complete temperature cycles. One cycle shall consist of maintaining the cylinder at minus 65 ±5° Fahrenheit for two hours, and the second cycle shall consist of maintaining the cylinder at plus 160 ±5° Fahrenheit for two hours. The cylinder shall then pass the requirements of 3.4.4. After completion of the temperature cycles, the cylinder shall

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TABLE II

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE CYLINDER

DEFECTS	MINOR	CRITICAL
Incorrect outlet configuration		X
Incorrect, incomplete, or faulty finish	X	
Incorrect type or class		X
Flaw in the material		X
Incorrect or incomplete marking	X	
Incorrect color	X	
Abnormally distorted or nonuniform weld		X
Loose particles inside the cylinder		X
Rust or scale		X
Irregularities in the internal surface greater than 1/16 inch		X
Existence of any foreign material, solid or liquid		X

TABLE III

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECTS
Exterior and interior markings	Missing incorrect, incomplete, illegible, of improper size, location, sequence, or method of application, markings not the same on the interior and exterior containers

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TABLE III (Cont'd)

ITEM	DEFECTS
Packaging and packing materials	Any nonconforming component, any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc ; bulging or distortion of the containers
Exterior and interior weight or content	Number per container is more or less than required, gross or net weight exceeds the requirements

be tested for leakage in accordance with 4.6 3 This test shall be performed before plating and marking.

4.6.5 Hydrostatic pressure - The cylinder shall be hydrostatically pressurized to an internal pressure of not less than 8000 psig at plus 70° Fahrenheit for a period of not less than 5 seconds Pressure shall be applied at the rate of 25,000 ±10,000 psi per minute. The cylinder shall pass the requirements specified in 3.4.5. This test shall be conducted after the cylinders have been marked

4 6 6 Rupture - The cylinder shall be subjected to gradual heating in air until ruptured The cylinder shall pass the requirement specified in 3.4.6.

4.6.7 Maximum filling density - The charged cylinder shall be weighed, then the nitrogen gas discharged and the cylinder weighed again. The difference between the empty and filled cylinder represents the quantity of nitrogen gas inside The cylinder shall then be filled with water and weighed to determine the weight of water The cylinder shall pass the requirement specified in 3.4.7.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The cylinders shall be packaged in accordance with MIL-STD-794, Level A or C, as specified in the contract or order (see 6 2) When Level A is specified, cylinders shall be preserved in accordance with MIL-P-116, Method IC, without the use of contact preservatives and without cleaning by immersion in solvent Package each cylinder in a fiberboard box, PPP-B-636, weather-resistant class, closed and sealed as specified in the appendix of PPP-B-636.

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5.2 Packing - Packing shall conform to MIL-STD-794, Level A, B, or C, as specified in the contract or order (see 6.2). For Levels A and B the gross weight of wood boxes shall not exceed 200 pounds.

5.3 Marking - Marking shall be as specified in MIL-STD-129.

6. NOTES

6.1 Intended use - The cylinders are intended to provide a power source for the pneumatic rescue hand tool.

6.2 Ordering data - Procurement documents should specify

(a) Title, number, and date of this specification.

(b) Whether first article inspection is required and where the samples should be delivered (see 4.3.1).

(c) Name and address of first article inspection laboratory (see 4.3.1).

(d) Cylinders subjected to destructive tests (see Table I) shall not be shipped or considered as part of the contract or order.

(e) Selection of applicable levels of preservation, packaging, and packing.

(f) Applicable methods of cleaning and preservation.

(g) Items of data required (see 6.3).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION
MIL-C-81575(AS) Cylinder, Nitrogen, For Rescue and Tool

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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